# Introduction to Control Systems - Problem Bank Questions

## Question 1

You have been tasked with creating a control system for an automated crane system that transports heavy material across a production space. The controller will control the movement of the load to ensure that as the operator moves the load along the desired path, the load does not swing or deviate too far.

Answer the following questions:

1. Do you think the control system should be closed-loop or open-loop? Why?
2. What specifications do you think might be included in a design document for the control behavior?

**Suggested Answer:**

1. The control system should be closed-loop because despite being able to model and control the load based on a straightforward first-principles model, the system has a high likelihood of being subject to external disturbances including wear on the equipment, debris on the tracks, changes in the distribution of the load as it moves, etc.
2. a) Maximum desired deviation from the trajectory, b) Response time to a change in the desired path, c) Time required for the moving load to settle at a set point, d). etc

## Question 2

Think about the operation of a cruise control system in a modern vehicle.

Answer the following questions:

1. What does the error represent in a conventional cruise control system?
2. List some of the operating conditions that the controller should be able to compensate for.
3. What specifications do you think might be included in a design document for the control behavior?

**Suggested Answer:**

1. The error represents the difference between the set desired speed, and the actual speed of the vehicle.
2. A cruise controller would be designed to compensate for changes in the set speed, changes in the conditions on the road including slope of the road, traction, wind, etc., the current throttle position, the current gear the vehicle is in, etc.
3. Minimum acceleration of the vehicle, maximum peak and steady-state error, fuel efficiency, behavior at low and high speed, response to unforeseen disturbances such as loss of traction, etc.

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